



Master of Science in Computer Science Cloud Computing Curriculum

Serverless CRATOR: A serverless crawler for dark-web

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JADS Jheronimus
Academy
of Data Science



What is a web crawler?

A web crawler is an automatic software designed to systematically and accurately traverse the web



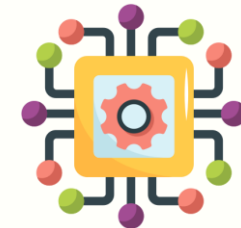
Dark web data is valuable for research



cybercrime analysis



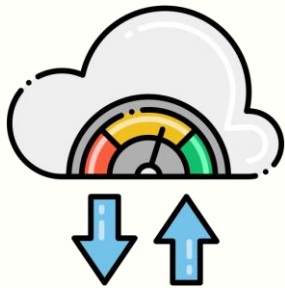
online market studies



threat intelligence

Applying these techniques to the dark web is significantly more challenging.

Challenges in Dark Web Crawling



high latency



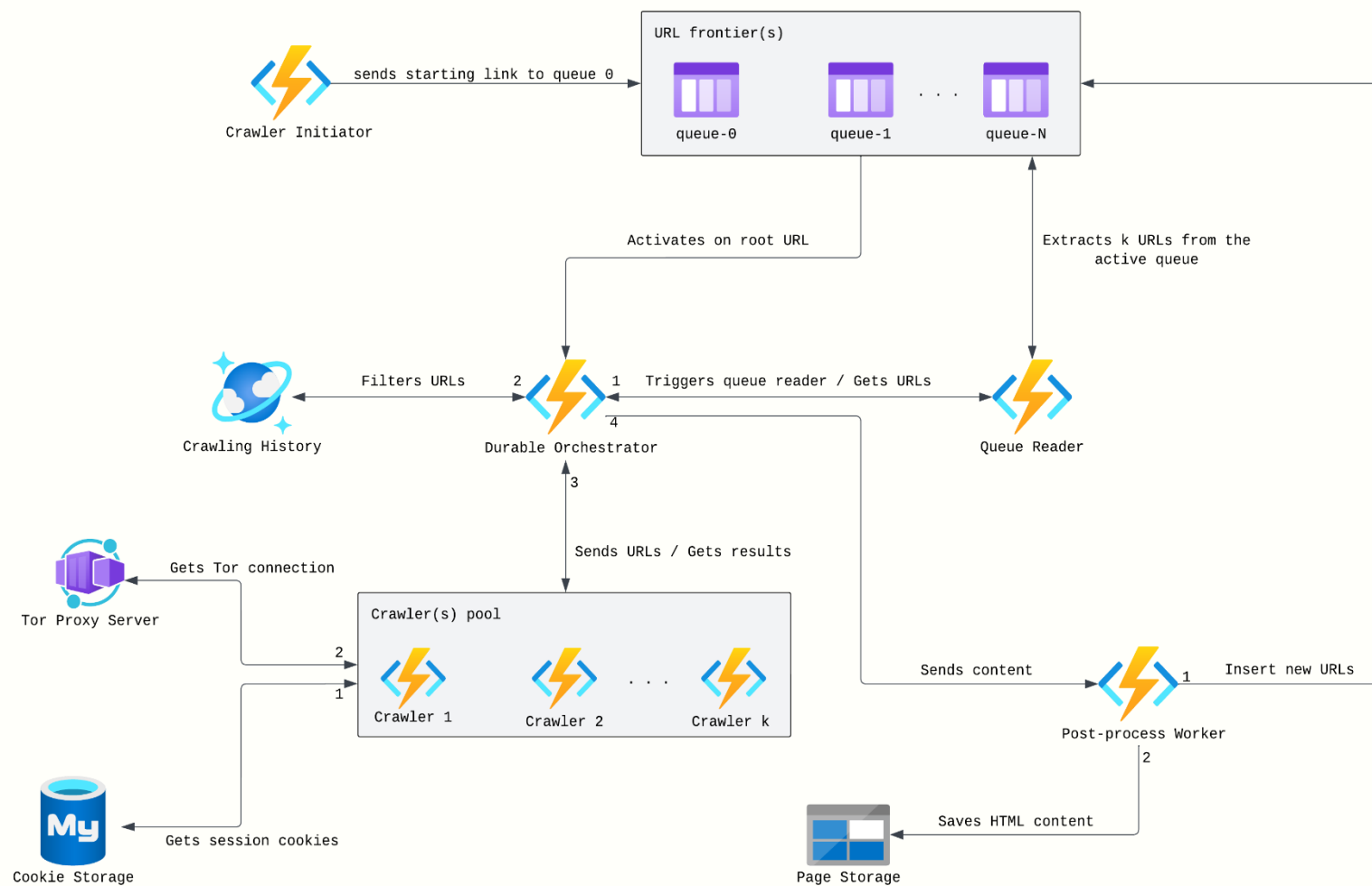
anonymity



services instability

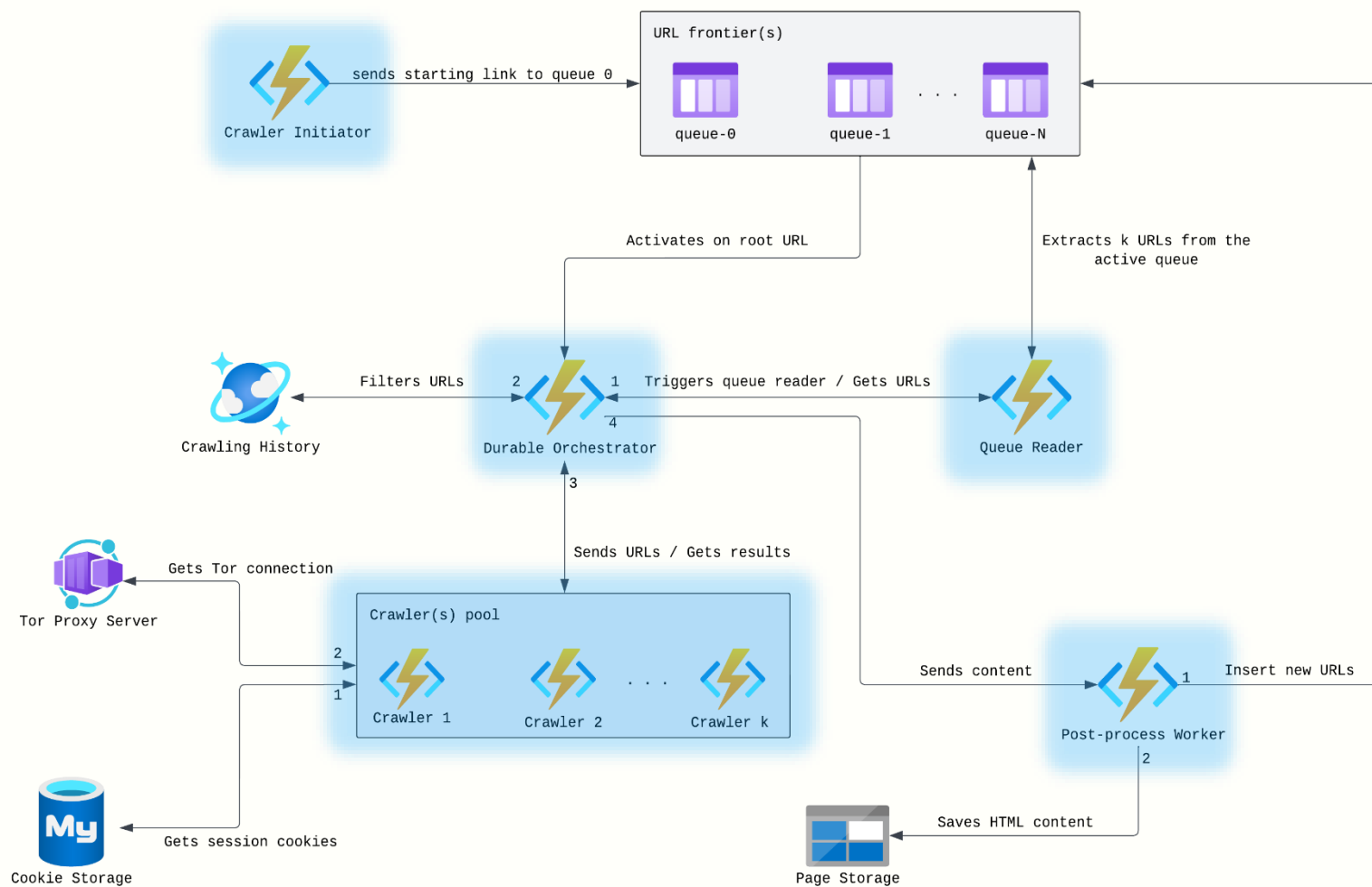
- RQ1.** Can a serverless architecture support dark web crawling?
- RQ2.** How does it compare with a traditional on-premise crawler?
- RQ3.** What are the trade-offs in terms of scalability, performance and cost?

Proposed Architecture



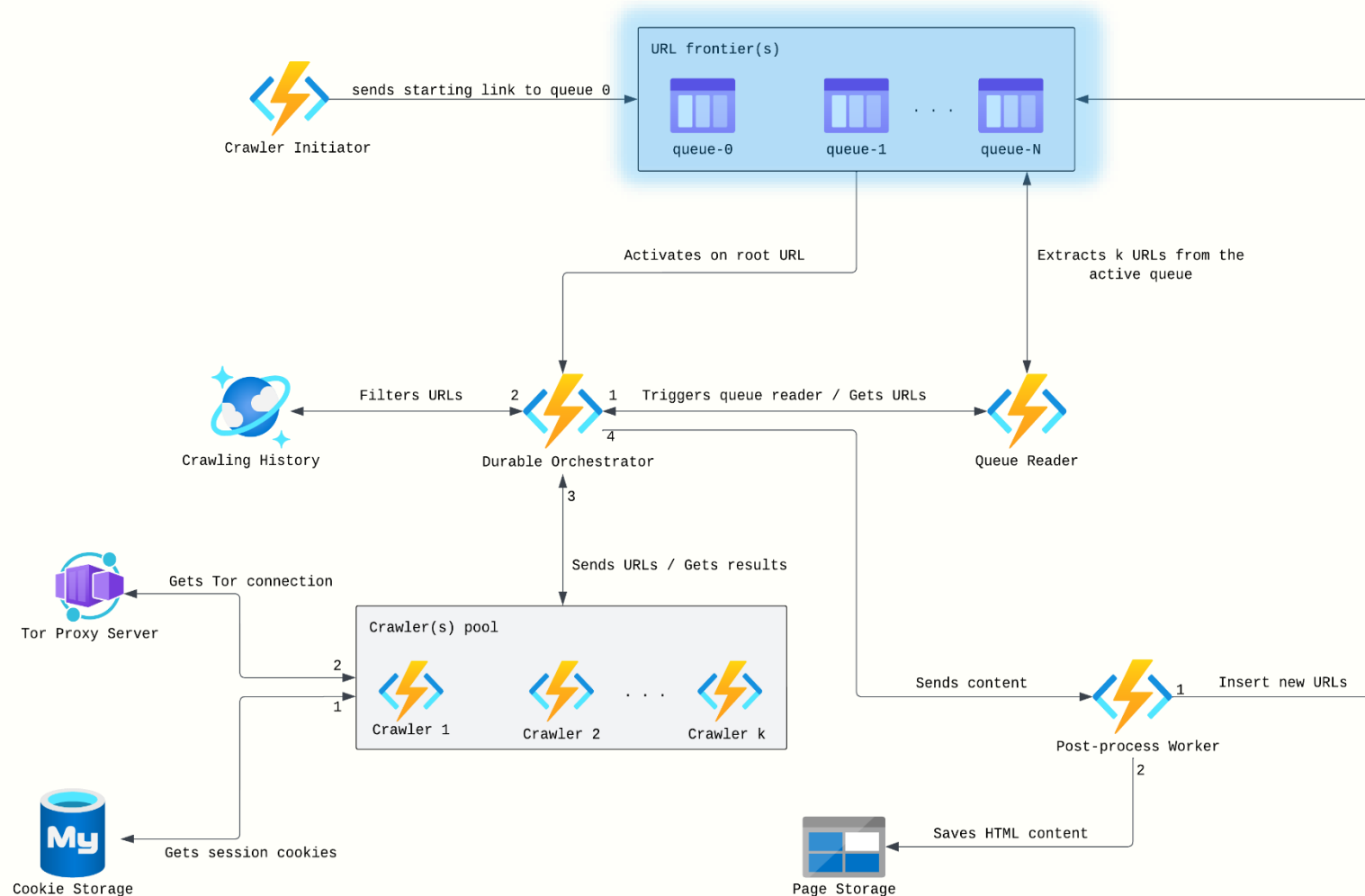
Microsoft Azure

Proposed Architecture



Durable Functions

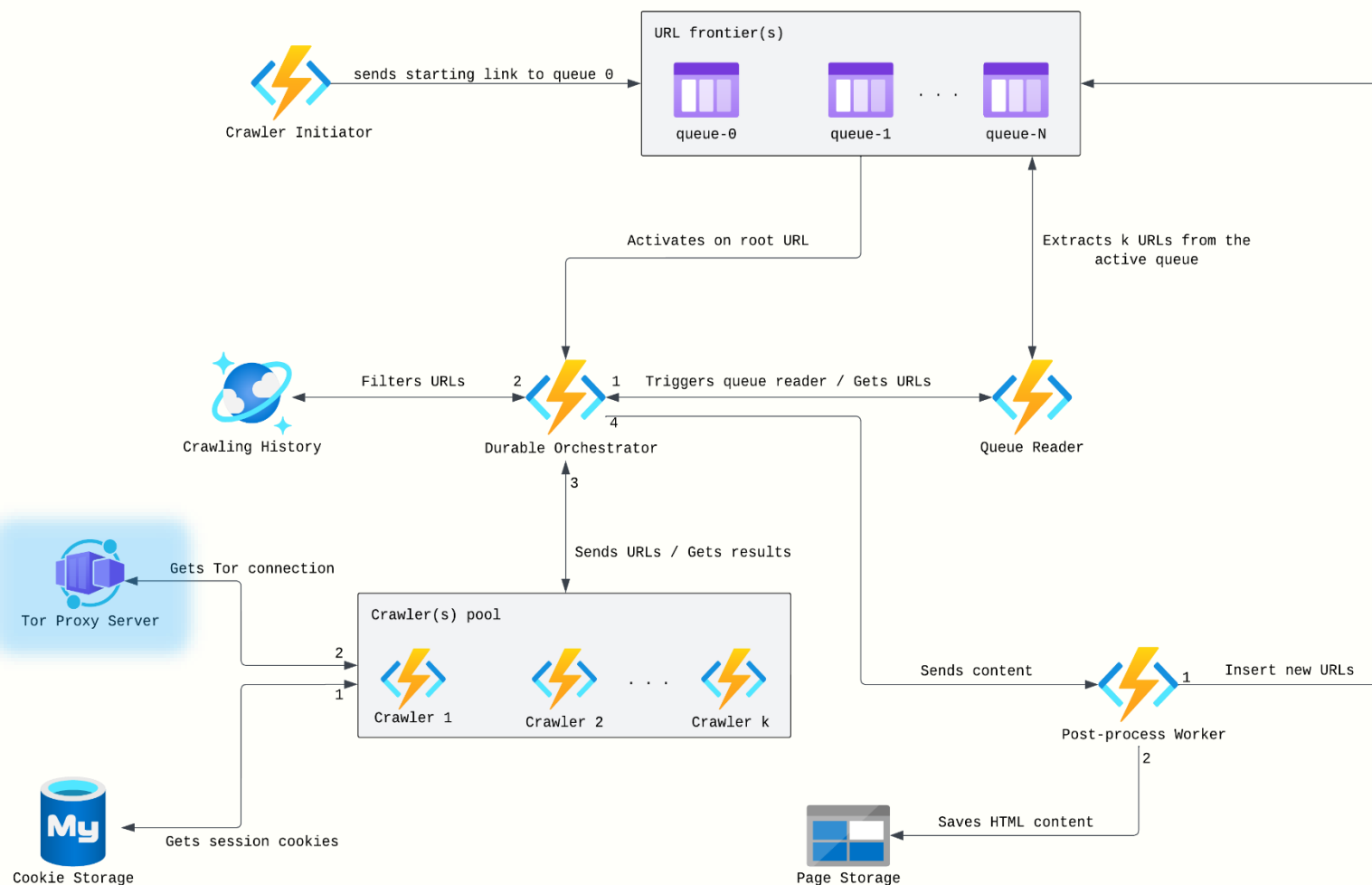
Proposed Architecture



Durable Functions

URL frontier

Proposed Architecture

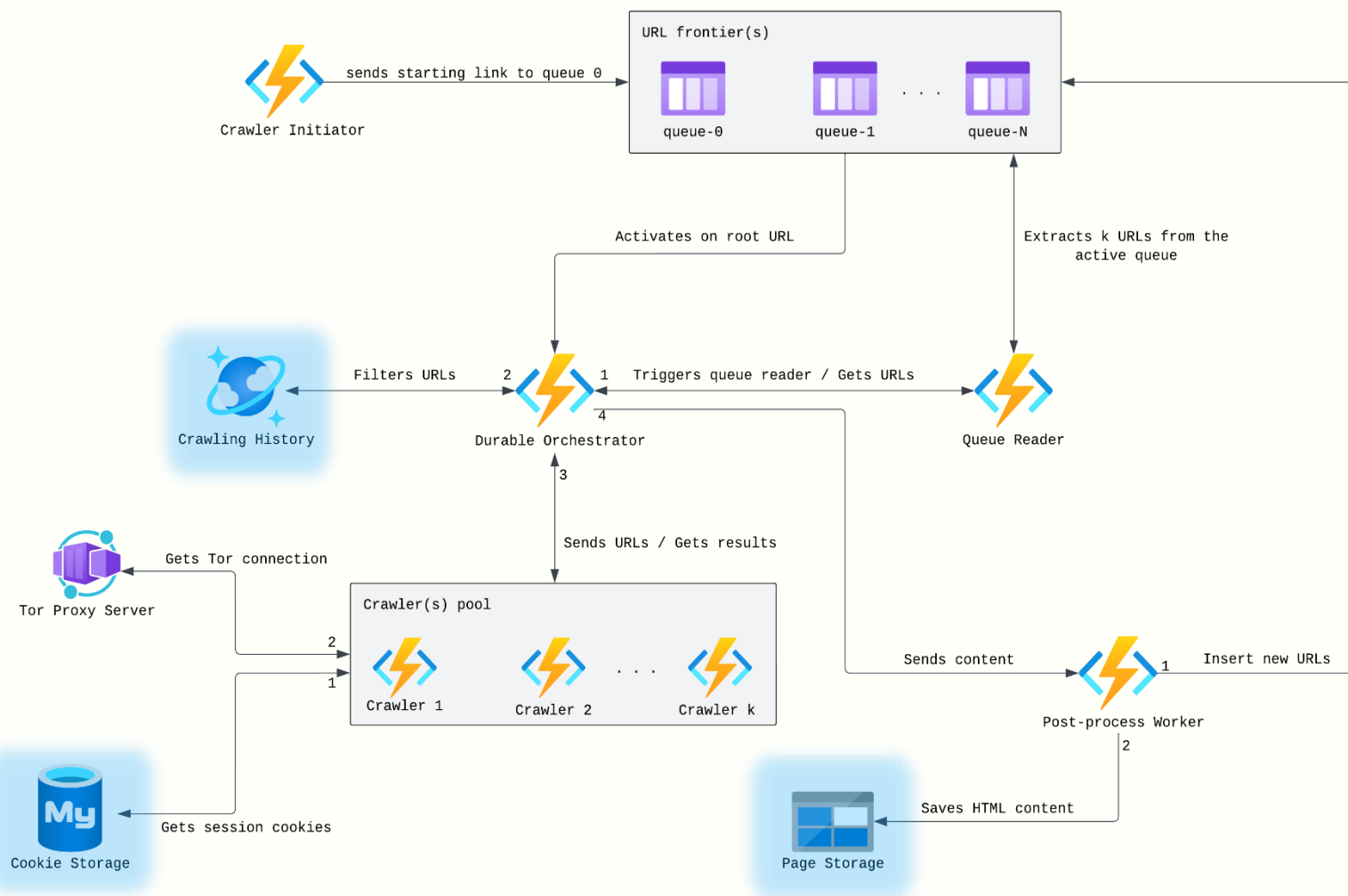


Durable Functions

URL frontier

Tor Proxy Container

Proposed Architecture

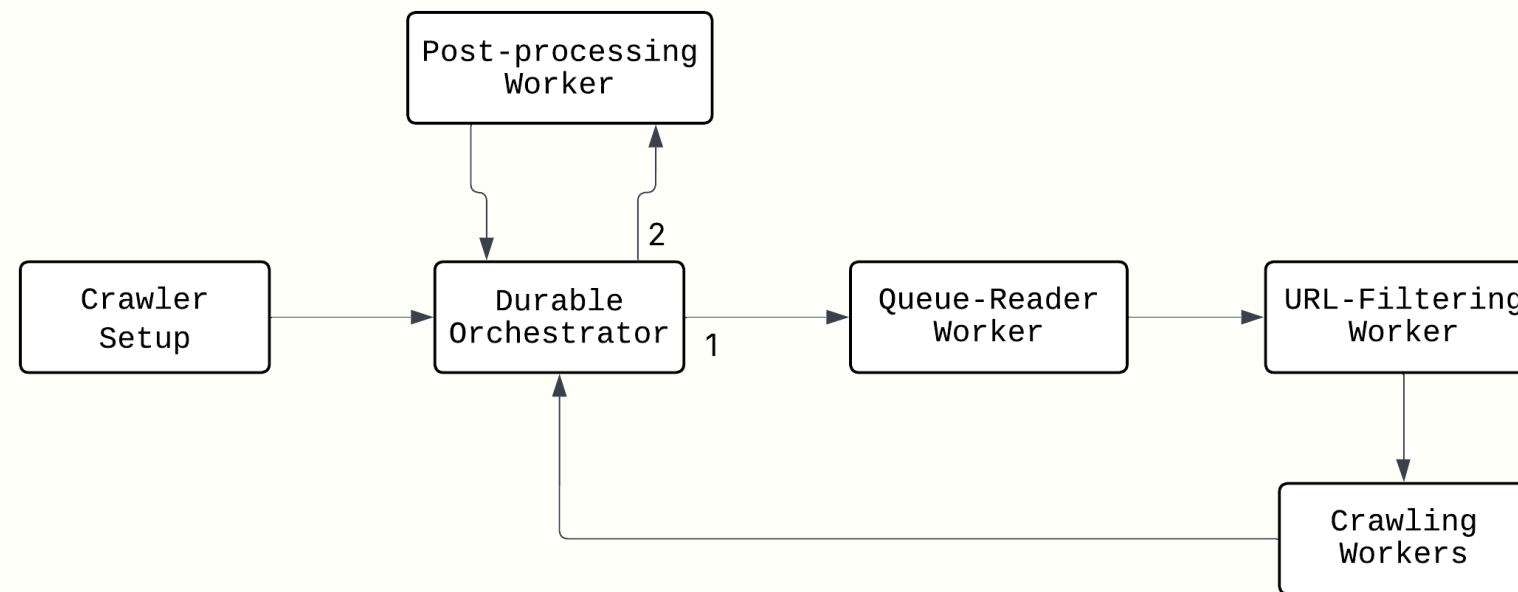


Durable Functions

URL frontier

Tor Proxy Container

Cloud Storage for data



Stop conditions

Empty Frontier Queue

Max Depth

Max Discovered URLs

Target Marketplace



Metrics for the evaluation

Coverage / Error Rate

Performance Metrics

Bulk Test (Immediate Reliability)

- High number of URLs in short time
- **Goal:** stress scalability and throughput



Temporal Test (Long-Term Stability)

- Crawling over extended time
- **Goal:** evaluate robustness and fault tolerance



Bulk Test

Metric	Serverless (5 Workers)	Serverless (10 Workers)	On-Premise
Coverage	99%	99%	100%
Error Rate	0.004%	0.017%	0%
Execution Time	10540.1	7429.1	3420.6s
Pages per Minute	11.8557	15.8930	39.076

Temporal Test

Metric	Serverless (5 Workers)	Serverless (10 Workers)	On-Premise
Coverage	99%	99%	100%
Error Rate	0.003%	0.003%	0%
Execution Time	10448.8	7719.8	3420.6
Pages per Minute	12.1496	16.2149	39.076

Metric	Serverless	On-Premise
Total Cost (test period)	€23.23	-
Number of Runs	120	120
Average Cost per run	€0.19/run	
Average Daily Cost	€0.59/day	-
Infrastructure Cost	Cloud Managed	€420 - €760 hardware
VPN Cost	Included in Infrastructure	€ 2.99/month
Cost-Model	Pay-per-Use (OpEx)	Up-front Investment (CapEx)
Break-Even Point	-	≈ 2400 - 3800 runs

Break-Even point will be reached after 6 - 10 years!

Serverless architectures can support dark web crawling

Comparable Coverage



Poor Execution Time



Strong Scalability



For researchers

- Short-term Analysis and Exploratory Research.
- Rapid Experimentation.
- Large-scale dark web data collection.

For Law Enforcement and Security Agencies

- Adaptive monitoring of emerging threats.
- Easier deployment of investigation tools.
- Threat intelligence gathering from hidden services.

For General Users

- Low entry barrier to experiment with dark web technologies.
- Suitable for educational or exploratory purposes.

Integration of automated content analysis

Adaptive scaling strategies for crawling workers

Support for advanced crawling strategies

Optimization of cloud resource usage

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Thank you all for your attention!